

# Supplementary Material: Within- and between-year seasonal patterns of respiratory pathogens and the potential for concurrent outbreaks

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*Keywords:* respiratory pathogens, influenza, RSV, seasonal dynamics, survival analysis, concurrent outbreaks, geographic variation

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## 1. Supplemental Materials

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## 1.1. Tables

### 1.1.1. Comparable Models by City and Pathogen

Table 1: Table S1: Number of comparable models ( $\Delta\text{AIC} < 7$ ) per city and pathogen. This table illustrates the distribution of model availability across the eight study cities, highlighting the variation in the number of comparable models per city.

| City               | Influenza     | RSV           |
|--------------------|---------------|---------------|
| <i>North</i>       |               |               |
| Beijing            | 20            | 39            |
| Lanzhou            | 2             | 4             |
| Xian               | 2             | 3             |
| <i>North Total</i> | <del>24</del> | <del>46</del> |
| <i>South</i>       |               |               |
| Wuhan              | 20            | 22            |
| Suzhou             | 18            | 3             |
| Guangzhou          | 4             | 11            |
| Wenzhou            | 2             | 2             |
| Yunfu              | 2             | 2             |
| <i>South Total</i> | <del>46</del> | <del>40</del> |
| <b>Grand Total</b> | <b>70</b>     | <b>86</b>     |

### 1.1.2. Median Probability of Concurrent Outbreaks by Risk Threshold

Table 2: Table S2: Median probability of concurrent outbreaks by city and risk threshold. Values represent the median proportion of years with concurrent high-risk periods for both Influenza and RSV across 1000 simulation paths. Risk thresholds are defined as joint top 30%, 20%, and 10% percentiles of disease burden.

| City         | Top 30% | Top 20% | Top 10% |
|--------------|---------|---------|---------|
| <i>North</i> |         |         |         |
| Beijing      | 0.200   | 0.100   | 0.000   |
| Lanzhou      | 0.118   | 0.059   | 0.000   |
| Xian         | 0.450   | 0.350   | 0.150   |
| <i>South</i> |         |         |         |
| Wuhan        | 0.400   | 0.300   | 0.150   |
| Suzhou       | 0.350   | 0.200   | 0.100   |
| Guangzhou    | 0.188   | 0.125   | 0.062   |
| Wenzhou      | 0.812   | 0.500   | 0.250   |
| Yunfu        | 0.375   | 0.188   | 0.125   |

### 1.1.3. Median Time to Next Concurrent Outbreak by Risk Threshold

Table 3: Table S3: Median time to subsequent concurrent outbreak (years) by city and risk threshold, with 95% confidence intervals. Values represent the x-intercepts of the dashed vertical lines in Figure 6, indicating the time (in years) at which 50% of simulation paths are expected to experience their subsequent concurrent outbreak. Confidence intervals (in parentheses) reflect uncertainty in the Kaplan-Meier survival estimates, which properly account for right-censoring of paths that never experience a second outbreak. The relatively narrow confidence intervals reflect high precision from aggregating across 1000 simulation paths per city. Values marked as “—” indicate that the 50% threshold was not reached within the simulation window. Risk thresholds are defined as joint top 30%, 20%, and 10% percentiles of disease burden.

| City         | Top 30%       | Top 20%       | Top 10%       |
|--------------|---------------|---------------|---------------|
| <i>North</i> |               |               |               |
| Beijing      | 3.0 (2.0–3.0) | 5.0 (4.0–6.0) | —             |
| Lanzhou      | 5.0 (5.0–6.0) | —             | —             |
| Xian         | 1.0 (1.0–1.0) | 2.0 (2.0–2.0) | 3.0 (3.0–3.0) |
| <i>South</i> |               |               |               |
| Wuhan        | 1.0 (1.0–1.0) | 2.0 (2.0–2.0) | 3.0 (3.0–3.0) |
| Suzhou       | 2.0 (2.0–2.0) | 3.0 (3.0–3.0) | 6.0 (5.0–6.0) |
| Guangzhou    | 3.0 (3.0–4.0) | 6.0 (6.0–7.0) | —             |
| Wenzhou      | 1.0 (1.0–1.0) | 1.0 (1.0–1.0) | 2.0 (2.0–2.0) |
| Yunfu        | 2.0 (2.0–2.0) | 3.0 (3.0–3.0) | 7.0 (6.0–8.0) |

#### 1.1.4. Optimal Within-Year and Between-Year Cycle Lengths

Table 4: Table S4: Optimal within-year and between-year cycle lengths (months) for each city and pathogen combination. Values represent the cycle lengths of the best-performing model (lowest AIC) for each city-pathogen pair. Within-year cycles are shown for both pathogens; between-year cycles are only shown for models that included multi-seasonal decomposition (MSTL). These values correspond to the data presented in Figure 3 (A) and (B) of the main text.

| Region | City      | Within-year cycle (months) |      | Between-year cycle (months) |      |
|--------|-----------|----------------------------|------|-----------------------------|------|
|        |           | Influenza                  | RSV  | Influenza                   | RSV  |
| North  |           |                            |      |                             |      |
|        | Beijing   | 13.0                       | 10.0 | 28.0                        | 24.0 |
|        | Lanzhou   | 13.0                       | 13.0 | 20.0                        | 35.0 |
|        | Xian      | 13.0                       | 13.0 | 36.0                        | 33.0 |
| South  |           |                            |      |                             |      |
|        | Suzhou    | 14.0                       | 12.0 | 26.0                        | 21.0 |
|        | Wuhan     | 13.0                       | 12.0 | 30.0                        | 19.0 |
|        | Wenzhou   | 12.0                       | 12.0 | 22.0                        | 18.0 |
|        | Guangzhou | 13.0                       | 12.0 | 34.0                        | 31.0 |
|        | Yunfu     | 14.0                       | 10.0 | 21.0                        | 24.0 |

## 1.2. Figures

### 1.2.1. Distribution of Between-Year Cycle Lengths

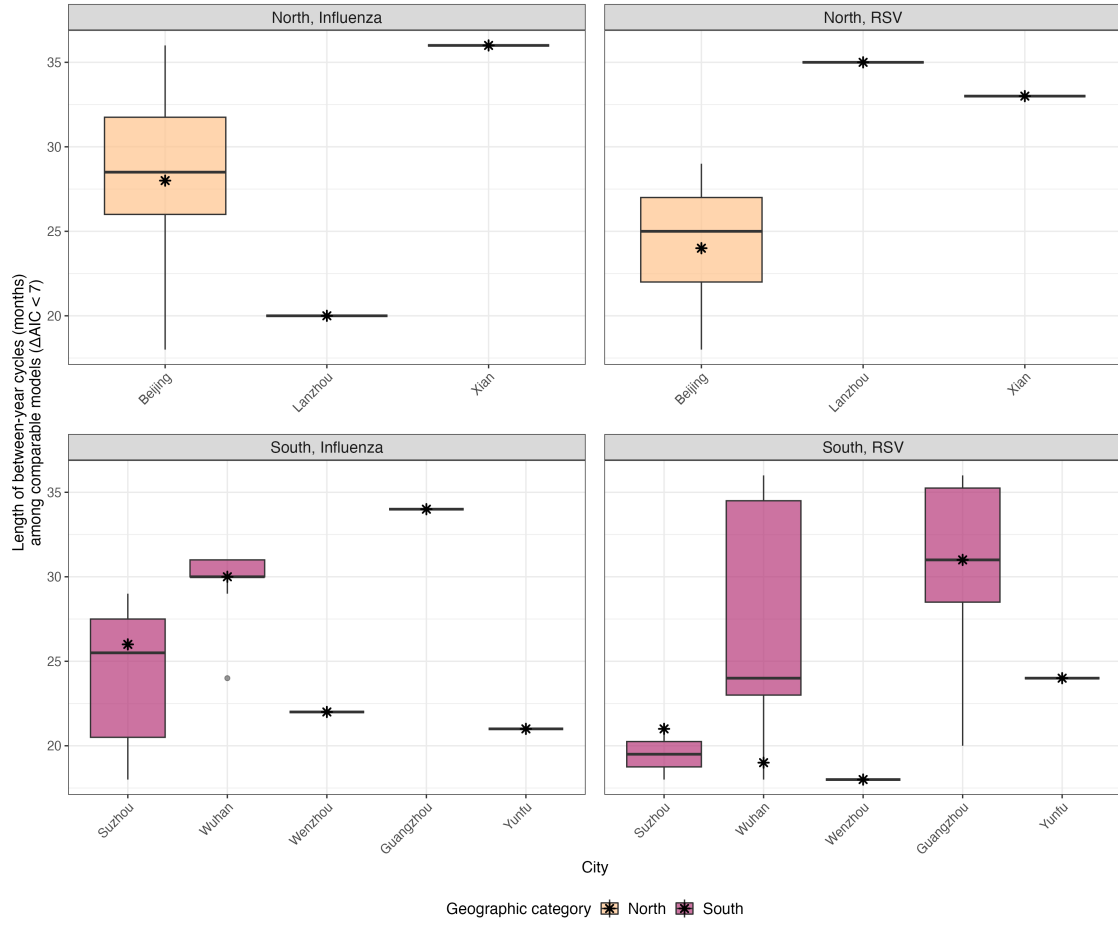


Figure 1: Figure S1: Distribution of between-year cycle lengths (months) among comparable models ( $\Delta AIC < 7$ ) by city and pathogen, separated by geographic region (North vs South). Each panel displays boxplots of the between-year cycle lengths (in months) for each city within the specified region and pathogen combination. The asterisk (\*) marks the best-performing model (lowest AIC) for each city-pathogen combination.

## Supplementary Text 1. Literature search strategy

We identified city-level surveillance data for respiratory syncytial virus (RSV) and influenza virus (InfV) through structured searches in PubMed. Searches were conducted on 12 August 2024, with no language restrictions. The full search strings and screening criteria are provided below.

### *RSV search*

The PubMed search string was:

```
("Respiratory Syncytial Viruses"[Mesh] OR RSV[Title/Abstract]  
OR "respiratory syncytial"[Title/Abstract])  
AND (China[Mesh] OR China[Title/Abstract] OR Chinese[Title/Abstract])
```

Studies were included if they met all of the following criteria:

- Surveillance data reported at the prefecture- or city-level (administrative level two);
- At least five consecutive years of observations;
- Monthly or weekly temporal resolution;
- Laboratory-confirmed RSV indicators (positivity among ARI/LRTI inpatients or outpatients).

Through this process, eight cities were identified with eligible RSV time series: Beijing, Lanzhou, Xi'an, Wuhan, Suzhou, Wenzhou, Guangzhou, and Yunfu.

### *Influenza search*

For the same eight cities, we applied a parallel PubMed search using the following query:

```
("Influenza, Human"[Mesh] OR influenza[Title/Abstract])  
AND (China[Mesh] OR China[Title/Abstract] OR Chinese[Title/Abstract])  
AND "last 10 years"[PDat]
```

The same inclusion criteria were applied (prefecture-level data,  $\geq 5$  years of continuous observations, and monthly or weekly reporting). For cities with multiple eligible influenza series, we selected the dataset with the longest uninterrupted coverage and the most specific virological definition. Data sources corresponding to each city-pathogen pair are listed in Supplementary Table S1.

Table 5: Supplementary Table S1. Surveillance indicator definitions for RSV and influenza (InfV)

| City      | Pathogen | Indicator definition                        | Clinical setting          | Laboratory      |
|-----------|----------|---------------------------------------------|---------------------------|-----------------|
| Beijing   | RSV      | RSV antigen positivity in ARI inpatients    | Inpatient (ARI)           | IFA/IFA         |
| Beijing   | InfV     | Influenza-positive samples by subtype       | Outpatient (ILI sentinel) | Virus isolation |
| Xi'an     | RSV      | RSV-positive cases / positivity rate        | Out-/inpatient (ARI)      | Not reported    |
| Xi'an     | InfV     | Daily ILI case counts (syndromic)           | Outpatient (ILI)          | None            |
| Lanzhou   | RSV      | RSV positivity; RSV-A genotypes             | Inpatient (ARI)           | RT-PCR          |
| Lanzhou   | InfV     | Laboratory-confirmed influenza              | Out-/inpatient            | RT-PCR          |
| Wuhan     | RSV      | RSV positivity in ARTI inpatients           | Inpatient (ARTI)          | DFA             |
| Wuhan     | InfV     | ILI% and influenza-positive specimens       | Outpatient (ILI sentinel) | RT-PCR          |
| Wenzhou   | RSV      | RSV-positive LRTI inpatients                | Inpatient (LRTI)          | DFA             |
| Wenzhou   | InfV     | Influenza-positive LRTI inpatients          | Inpatient (LRTI)          | DFA             |
| Suzhou    | RSV      | RSV positivity in neonatal LRTI inpatients  | Inpatient (neonatal LRTI) | DFA             |
| Suzhou    | InfV     | Influenza positivity in SARI inpatients <5y | Inpatient (SARI)          | RT-PCR          |
| Guangzhou | RSV      | RSV positivity in ARI patients              | In-/outpatient (ARI)      | qRT-PCR         |
| Guangzhou | InfV     | ILI%, influenza positivity, ILI+ proxy      | Outpatient (ILI)          | RT-PCR          |
| Yunfu     | RSV      | RSV positivity in hospitalized children     | Inpatient (pediatric ARI) | DFA             |
| Yunfu     | InfV     | Influenza positivity in ILI outpatients     | Outpatient (ILI sentinel) | Isolation       |

<sup>a</sup> Cui G et al. *PLoS One*. 2013.<sup>b</sup> Wu S et al. *Influenza Other Respir Viruses*. 2018.<sup>c</sup> Zhang L et al. *Modern Preventive Medicine*. 2018.<sup>d</sup> Chen D et al. *Entropy*. 2023.<sup>e</sup> Liang X et al. *Medicine*. 2019.<sup>f</sup> Du M et al. *PLOS One*. 2022.<sup>g</sup> Hu X-W et al. *JMIR Public Health Surveill*. 2023.<sup>h</sup> Liu X et al. *BMC Infect Dis*. 2023.<sup>i</sup> Lin J. Master's thesis, Wenzhou Medical University. 2011.<sup>j</sup> Zhong P-P et al. *Chin J Contemp Pediatr*. 2016.<sup>k</sup> Lu L et al. *BMC Infect Dis*. 2015.<sup>l</sup> Yu J et al. *Pediatr Infect Dis J*. 2019.<sup>m</sup> Zou L et al. *Front Microbiol*. 2016.<sup>n</sup> Yan Z-L et al. *BMC Public Health*. 2024.<sup>o</sup> Qin C et al. *China Practical Medicine*. 2022.<sup>p</sup> Peng L-X et al. *Practical Preventive Medicine*. 2016.